

		answer
2(a)(i)	The rotor blades push the air downwards, causing air particles to <u>experience a change in momentum in the downward direction</u> , hence blades <u>exert a downward force on air particles (by Newton's second law)</u> . According to <u>Newton's third law</u> , the <u>air exerts an equal and opposite upward force/push (thrust)</u> on the blades.	[1] [1]
2(a)(ii)	$\frac{dV}{dt} = A \frac{d(h)}{dt}$ $\frac{dV}{dt} = Av$ $\frac{dV}{dt} = (\pi \times 5.5^2)(18)$ $= 1711$ $\approx 1700 \text{ m}^3 \text{ s}^{-1}$	[1] correct expression and substitution [1] correct ans
2(a)(iii)	By Newton's third law, force acting on air by blades has the same magnitude as force acting on blades by air Applying $F = \Delta v \frac{dm}{dt}$ force on air due to blades = $\Delta v \frac{dm}{dt}$ $F = \Delta v \frac{dm}{dt}$ $F = \Delta v \rho \frac{dV}{dt}$ $= 18 \times 1.02 \times 1711$ $= 31414$ $\approx 31400 \text{ N}$	[1] correct expression and substitution [1] correct ans

No.	Solution	Remarks
2(b)(i)	Impulse is the <u>product of (average) force and time which the force acts</u> . To change <u>the momentum of the package from zero to a downward value</u> , by Newton's second law, a net force in the downward direction must act on package over the time interval.	[1] correct definition of impulse [1] correct explanation
2(b)(ii)	$\langle F \rangle \Delta t = \Delta p$ $\langle F \rangle = \frac{p_f - p_i}{\Delta t}$ $\langle F \rangle = \frac{mv - 0}{\Delta t} = \frac{50(4.0) - 0}{0.20} = 1000 \text{ N}$	[1] correct expression and substitution [1] correct ans